

# Action Differences, Fluctuations and Operational Bounds

Control results for a research programme on mechanical gaps

navstokgap research working notes

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## Abstract

For a particle launched perpendicular to a constant force, we derive the exact chord-curve action difference  $\Delta S = F^2 T^3 / (24m) = T \delta E / 12$  and relate it to the geometric error area. Positive fixed-endpoint action differences accumulate at zero; a Taylor argument extends this result to nondegenerate variations. For constant force, the Dirichlet Jacobi operator has a positive lowest eigenvalue, separating normalized fluctuation cost from action amplitude. The free quantum kernel converges to a delta with a second-derivative short-time correction. Finally, within a specified quantum two-arm measurement model, we derive an exact action-resolution threshold for  $N$  independent copies and target success probability  $p$ . For fixed  $1/2 < p < 1$ , the threshold decreases as  $N^{-1/2}$ . These results connect trajectory geometry, fluctuation spectra and operational resolution in a tractable setting for studying gap formation.

## 1 Action values, fluctuations and resolution

The motivating question is whether limitations on Newtonian trajectories can explain a nonzero action scale, and whether a useful toy model can isolate the formation of a gap. We write  $h = 2\pi\hbar$  for Planck's constants,  $\eta$  for a path variation, and  $J$  for a second-variation operator. An additional field  $\chi$  would enter through a specified extended action  $S[q, \chi]$ .

For a path class  $\mathcal{P}$  and reference  $q_*$ , one possible action-gap question is

$$g_S = \inf\{|S[q] - S[q_*]| : q \in \mathcal{P}, S[q] \neq S[q_*]\} > 0.$$

Assume the set is nonempty and specify the path class, topology, endpoints and duration. The corresponding Hamiltonian gap question is  $\text{spec}(H) \cap (E_0, E_0 + \Delta) = \emptyset$  for some  $\Delta > 0$ . Here  $E_0$  is the ground energy of a specified self-adjoint Hamiltonian. The path-space infimum and spectral interval give two precise meanings of a gap.

Newton's geometry motivates the area comparison [7, 8]. NATP00385 [6] supplies Newton's later account of analytic discovery and synthetic presentation. The following calculation expresses the geometric comparison through Hamilton's action.

## 2 Constant force: geometry and action

Fix  $m, F, T, v_0 > 0$ , set  $V(y) = -Fy$ , and impose  $x(0) = y(0) = \dot{y}(0) = 0, \dot{x}(0) = v_0$ . Newton's equations give

$$x_*(t) = v_0 t, \quad y_*(t) = \frac{F t^2}{2m}, \quad 0 \leq t \leq T.$$

The kinetic-energy gain is  $\delta E = F^2 T^2 / (2m) = -\Delta V$ ; total mechanical energy is conserved. The triangle between initial tangent, endpoint vertical and chord has area  $v_0 F T^3 / (4m)$ . The

chord–curve lens has one third of this:

$$\mathcal{A}_{\text{lens}} = \frac{v_0 F T^3}{12m}.$$

These are frame-dependent areas in configuration space. The factor relating the lens area to action will follow from a matched-endpoint variation.

Hold  $x = x_*$  fixed and consider the same-endpoint paths  $y = y_* + \eta$  with  $\eta \in H_0^1(0, T)$ , the Sobolev space with square-integrable weak derivative and zero endpoint trace. Define

$$S[x, y] = \int_0^T \left[ \frac{m}{2} (\dot{x}^2 + \dot{y}^2) + Fy \right] dt.$$

The linear variation vanishes by integration by parts and  $m\ddot{y}_* = F$ :

$$S[x_*, y_* + \eta] - S[x_*, y_*] = \frac{m}{2} \int_0^T \dot{\eta}^2 dt. \quad (1)$$

The straight chord  $y_{\text{ch}} = FTt/(2m)$  is an admissible off-shell comparison path. Substitution into (1) gives

$$\Delta S_{\text{ch},*} = \frac{F^2 T^3}{24m} = \frac{F}{2v_0} \mathcal{A}_{\text{lens}} = \frac{T\delta E}{12}. \quad (2)$$

Adding the same total time derivative  $dG(q, t)/dt$  to both Lagrangians leaves this difference unchanged because configuration endpoints and duration coincide. A velocity-dependent endpoint function requires additional endpoint data.

**Proposition 1** (Zero infimum of action differences). *For this fixed-endpoint path class the infimum of positive action differences from the classical path is zero, although the classical path is the unique minimizer within the class with fixed horizontal motion.*

*Proof.* If  $\eta \neq 0$  in  $H_0^1$ , its derivative cannot vanish almost everywhere, so (1) is positive. For  $\eta_a(t) = at(T - t)$  it equals

$$\Delta S(a) = \frac{ma^2 T^3}{6}.$$

Any prescribed positive bound is violated by taking a sufficiently small nonzero  $a$ . The zero variation alone gives equality in (1).  $\square$

**Proposition 2** (Local obstruction along a variation). *Let  $q_a$  be admissible for all  $a$  in an open interval about zero, and suppose  $f(a) = S[q_a]$  is twice continuously differentiable there, with  $f'(0) = 0$  and  $f''(0) = b \neq 0$ . Then arbitrarily small nonzero absolute action differences  $|S[q_a] - S[q_0]|$  occur. If  $b > 0$ , these differences can be taken positive without absolute values.*

*Proof.* Taylor's formula gives  $f(a) - f(0) = ba^2/2 + o(a^2)$ . For all sufficiently small nonzero  $a$ , its magnitude lies between  $|b|a^2/4$  and  $3|b|a^2/4$  and its sign is the sign of  $b$ . It is nonzero and tends to zero.  $\square$

Locality here is initially in the parameter  $a$ . To conclude that these actions occur in every path-space neighbourhood of  $q_0$ , additionally require  $q_a \rightarrow q_0$  in the specified path topology. The polynomial family above converges in  $H^1$  (and in  $C^1$ ).

The admissible small-amplitude family is the mechanism behind both propositions. Discrete path classes, topological sectors and operational equivalence relations offer concrete ways to change this admissibility question. Kepler collisions pose a separate IVP question about continuation through a singular force.

### 3 Jacobi fields and normalized fluctuation gaps

For  $S[q] = \int_0^T (m\dot{q}^2/2 - V(q))dt$  about a smooth classical solution, the second variation is

$$Q[\eta] = \int_0^T [m\dot{\eta}^2 - V''(q_*)\eta^2] dt, \quad J = -m \frac{d^2}{dt^2} - V''(q_*).$$

For smooth bounded  $V''(q_*(t))$  take the Dirichlet operator domain  $H^2(0, T) \cap H_0^1(0, T)$  in  $L^2(0, T)$ . A variation through classical solutions satisfies the Jacobi equation  $J\eta = 0$ . The full variational path class also contains off-shell fluctuations. For constant force a Jacobi field is  $A + Bt$ , so fixed zero endpoints force it to vanish. Equation (1) nevertheless admits nonzero fluctuations.

The second-variation framework is classical [1, §8.2]. For constant force the Dirichlet eigenfunctions are  $\sin(n\pi t/T)$  with eigenvalues  $\lambda_n = m(n\pi/T)^2$ ,  $n \geq 1$ . Thus

$$Q[\eta] \geq \frac{m\pi^2}{T^2} \|\eta\|_{L^2}^2.$$

This is coercivity relative to the  $L^2$  norm: rescaling  $\eta$  rescales  $Q$  quadratically. The eigenvalue  $\lambda_n$  has units mass/time<sup>2</sup>, while  $\|\eta\|_{L^2}^2$  has units length<sup>2</sup> time, producing action.

For the oscillator  $V = m\omega^2 q^2/2$ , the corresponding eigenvalues become  $m[(n\pi/T)^2 - \omega^2]$ . Zero modes occur at  $T = n\pi/\omega$ ; positivity of all modes fails after the first such time. This explicit calculation is a useful comparison with the quantum oscillator Hamiltonian. The companion working draft *A Gap Laboratory: Oscillators, Circles and Winding Sectors* derives both spectra, their domains and their distinct gap-closing limits.

### 4 What distribution is the oscillatory kernel?

The quantum path-phase input is  $\exp(iS/\hbar)$ , with time-integrated action, as in the Feynman formulation [3, 4]. To study its distributional meaning, specify a kernel, normalization and limiting parameter. The free one-dimensional Schrödinger kernel provides an exact example [9, §7.3, (7.30)]. The expansion below is its Fourier consequence. For  $\hbar, m, \tau > 0$  let

$$K_\tau(z) = \left( \frac{m}{2\pi i \hbar \tau} \right)^{1/2} \exp \left( \frac{imz^2}{2\hbar\tau} \right),$$

with the usual branch  $1/\sqrt{i} = e^{-i\pi/4}$  and oscillatory/tempered-distribution interpretation. Use Fourier transform  $\widehat{f}(k) = \int e^{-ikz} f(z) dz$ . Equivalently this kernel is defined by

$$\widehat{K}_\tau(k) = \exp \left( -\frac{i\hbar\tau k^2}{2m} \right).$$

For each Schwartz test function, dominated convergence in Fourier space implies

$$K_\tau \longrightarrow \delta \quad (\tau \downarrow 0, \hbar > 0 \text{ fixed}). \quad (3)$$

Taylor expansion of the multiplier, using  $|e^{-iu} - 1 + iu| \leq u^2/2$  for real  $u$ , further gives

$$K_\tau = \delta + \frac{i\hbar\tau}{2m} \delta'' + O(\tau^2) \quad \text{in the sense of pairing with each fixed Schwartz test function.}$$

The remainder estimate follows by integrating  $k^4$  times the absolute value of the transformed test function. The even free kernel gives  $\delta$  at leading order and  $\delta''$  at first order in time. Differentiating the kernel with respect to  $z$  instead gives a family converging to  $\delta'$ . A Fourier constraint integral provides another construction:  $\int e^{i\lambda z/\hbar} d\lambda/(2\pi\hbar) = \delta(z)$ . Substitution of a nonlinear constraint requires further regularity and Jacobian conditions.

The limit above is short time at fixed  $\hbar$ . A semiclassical analysis instead varies  $\hbar$ ; each limit has its own test functions and convergence statement.

## 5 A conditional operational action threshold

Assume the quantum state/measurement framework, a specified positive  $\hbar$ , and a coherently controlled two-arm system realizing a relative phase  $\phi = \Delta S/\hbar$ . The relative arm phase encodes two known hypotheses with equal priors:

$$|\psi_0\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}, \quad |\psi_\phi\rangle = \frac{|0\rangle + e^{i\phi}|1\rangle}{\sqrt{2}}.$$

Allow  $N \geq 1$  independent identical copies and any joint binary quantum measurement. These assumptions define the resource budget and the hypothesis test.

The following is a pure-state specialization of the Holevo–Helstrom theorem [10, Theorem 3.4], followed by the specified action-phase encoding.

**Proposition 3** (Finite-copy bound). *The optimal equal-prior success probability is*

$$P_N(\phi) = \frac{1}{2} \left( 1 + \sqrt{1 - \cos^{2N}(\phi/2)} \right). \quad (4)$$

For  $1/2 < p \leq 1$ , in the first phase lobe  $|\phi| \leq \pi$ , success at least  $p$  is possible if and only if

$$|\Delta S| \geq d_{N,p} := 2\hbar \arccos \left( [4p(1-p)]^{1/(2N)} \right). \quad (5)$$

For each fixed  $1/2 < p < 1$ ,  $d_{N,p} \rightarrow 0$  as  $N \rightarrow \infty$ . For  $p = 1$ , the finite-copy threshold instead remains  $\pi\hbar$ .

*Proof.* Two normalized pure states with overlap magnitude  $r$  have density-matrix difference  $D$  with eigenvalues  $\pm\sqrt{1-r^2}$  on their span (and zero outside). Indeed  $\text{tr } D = 0$  and  $\text{tr } D^2 = 2(1-r^2)$ . A binary measurement effect  $0 \leq E \leq I$  gives success  $1/2 + \text{tr}(ED)/2$ , maximized by projecting onto the positive eigenspace. The maximum is  $(1 + \sqrt{1-r^2})/2$ . Here a single-copy overlap has magnitude  $|\cos(\phi/2)|$ , and tensor products raise it to the  $N$ th power, proving (4). Within the first lobe cosine is nonnegative and monotone in  $|\phi|$ , so solving  $P_N \geq p$  gives (5). For  $0 < 4p(1-p) < 1$  its  $1/(2N)$  power tends to one. At  $p = 1$  it is zero for every finite  $N$ , giving the stated exception.  $\square$

For example  $N = 1$ ,  $p = 3/4$  gives  $d_{1,3/4} = \pi\hbar/3$ . For fixed  $1/2 < p < 1$  the asymptotic threshold is

$$d_{N,p} \sim 2\hbar \sqrt{\frac{-\log(4p(1-p))}{N}}.$$

The threshold is positive at fixed resources and accuracy. Its dependence on  $N$  quantifies the improvement in resolution from repeated preparations. Phase periodicity explains the first-lobe restriction: at  $\phi = 2\pi n$  the hypotheses coincide again.

If an apparatus realizes the particular relative action (2), with all control and recombination phases included, its first-lobe threshold would be

$$T \geq \left( \frac{24md_{N,p}}{F^2} \right)^{1/3}.$$

The upper first-lobe condition  $F^2 T^3 / (24m\hbar) \leq \pi$  must be retained. Realizing the off-shell chord as an experimental arm requires a control force; the displayed substitution applies when the full apparatus has the specified relative action.

## 6 Finite propagation and the field-theory companion

For the free relativistic Lagrangian  $L = -mc^2\sqrt{1 - \dot{q}^2/c^2}$ , consider the fixed-endpoint family  $q_a = at(T - t)$ , with both endpoints zero. If  $|a|T < c$ , the path has a strict speed margin. Relative to the rest path its action excess is positive for  $a \neq 0$  and

$$\Delta S(a) = mc^2 \int_0^T \left(1 - \sqrt{1 - a^2(T - 2t)^2/c^2}\right) dt = \frac{ma^2T^3}{6} + O(a^4) \longrightarrow 0.$$

Thus positive action differences accumulate at zero within the strict speed margin, just as in the Newtonian example. This gives a kinematic comparison at finite  $c$ .

Navier–Stokes and Yang–Mills remain complementary comparisons, with their official targets kept intact [2, 5]. A diffusion or Dirichlet coercivity bound can depend on time, domain and normalization; a physical quantum mass gap concerns a different operator and limiting problem. Any future bridge must identify the spaces, physical versus auxiliary time, estimates and limits it preserves. The fluid target is regularity of evolution; the field-theory target includes construction of the physical quantum theory and its vacuum gap.

## 7 Scope, reproduction and next experiments

The results concern the stated mechanical path classes and quantum measurement model. Hilbert states, the Born rule and positive  $\hbar$  are physical inputs to Section 5. Deriving these premises belongs to the reconstruction branch of the companion programme. The proofs received internal review; novelty assessment and external review are publication-stage tasks.

The repository’s **make check** verifies the exact polynomial action identities, the oscillator-mode residual, the Fourier expansion coefficient and elementary discrimination cross-checks. **make papers** rebuilds this note and the companion programme PDF. The formalisation queue begins with the small-amplitude polynomial lemma.

The companion *Time Refinement and an Action Scale* develops the free Gaussian blocking test. Next steps audit the regulator proposal and quantum reconstruction axioms, alongside central-potential models and the historical corpus inventory, with sequential source audits. A successful extension must identify which premise excludes the classical small-amplitude families, or else choose an operational or spectral gap instead and state the change of target. Deriving a theory’s form, forcing a nonzero universal scale and determining its measured numerical value are separate tasks.

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